

David Kreuter

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Senior Network & Infrastructure Automation Engineer

Infrastructure engineer with 10+ years of experience building automated provisioning platforms, network orchestration systems, and API-driven workflows at enterprise scale. Architected a global provisioning framework that manages 20,000+ virtual machines across three geo-redundant datacenters, integrating VMware, Infoblox, Check Point, and Rubrik APIs into zero-touch deployment pipelines. Demonstrated expertise in network automation, security policy enforcement, IP lifecycle management, Python/PowerShell development, and infrastructure-as-code with Terraform. Proven track record of delivering compliant, high-availability infrastructure in regulated financial environments.

WORK EXPERIENCE

M&T Bank

03/2019 - Present

Automation Engineer (formerly Systems Engineer)

- **Network Security Automation:** Automated Check Point firewall rule generation and DMZ provisioning, integrating security policy enforcement into the server build cycle to ensure day-zero network isolation--directly analogous to SDN security group and policy management workflows.
- **Architectural Scale:** Engineered a global server provisioning framework that deploys and manages 20,000+ virtual machines; integrated Rubrik and VMware APIs to automate backup, compute placement, and network resource allocation, ensuring 100% RTO/RPO compliance at build time.
- **Network Provisioning & Lifecycle Management:** Developed an intelligent IP provisioning engine through Infoblox API integrations, handling subnet allocation, routing configuration, and network resource lifecycle; eliminated 1,000+ annual engineer hours and resolved a multi-year infrastructure bottleneck.
- **Platform Engineering & Middleware:** Designed and developed a custom Python orchestration application as middleware between ServiceNow and Ansible Automation Platform; implemented advanced retry logic, auto-remediation of deployment failures, and a pub/sub queue model for high-volume scaling.
- **Hybrid Cloud & GitOps:** Architected GitOps-driven CI/CD pipelines using Terraform to automate infrastructure deployment to Azure; implemented a pull-request-based workflow that standardized infrastructure changes through automated plan/apply cycles.
- **Infrastructure Resilience:** Engineered and maintain a highly resilient infrastructure spanning three identical, geo-redundant datacenters with isolated network segments to maximize uptime and ensure disaster recovery readiness.
- **Cloud Abstraction & Redesign:** Refactored legacy on-premises automation workflows to be fully cloud-agnostic, creating an abstraction layer that enables seamless deployment across on-premises and Azure environments while maintaining a consistent provisioning interface.
- **End-to-End Hardening & Provisioning:** Developed a zero-touch build workflow that delivers fully patched and hardened Linux (Ansible) and Windows (GPO) servers; automated agent installation, drive formatting, and security baseline application at scale.
- **Database as a Service (DBaaS):** Collaborated with Database Engineering to architect a high-performance on-premises SQL cluster, enabling a unified DBaaS offering that consolidated fragmented legacy databases and streamlined application delivery.
- **Cost Optimization:** Orchestrated the migration of 850 branch file servers within a strict three-month window using custom PowerShell automation, avoiding \$400,000 in extended licensing fees.
- **Risk Mitigation:** Authored Active Directory lifecycle scripts with built-in dry-run modes; implemented least-privilege permissions and robust rollback protocols for stale object cleanup across production environments.
- **Technical Leadership:** Serve as a primary mentor for the Technology Development Program (TDP), setting enterprise-wide standards for naming conventions, Git branching strategies, and unit testing for infrastructure code.

OTG Management

07/2013 - 02/2019

Lead Network Engineer

- **Infrastructure Resilience:** Architected a highly redundant DNS solution across a distributed network, significantly reducing outages and optimizing global traffic routing.
- **Cloud Connectivity:** Established secure, encrypted VPN tunnels to bridge on-premises networks with cloud resources, enabling seamless hybrid-cloud operations with proper network segmentation.
- **Performance Engineering:** Diagnosed and resolved a mission-critical server hang by identifying a NUMA architecture mismatch; realigned vCPU/memory topology to eliminate cross-node latency and restore system stability.
- **Security Architecture:** Engineered a secure reverse proxy solution to protect sensitive internal infrastructure, enabling customer-facing services while mitigating external threat vectors.

Network Engineer

- **Network Administration:** Managed complex firewall rules, switch ACLs, and VLAN configurations for a massive distributed network spanning 12 airports and 3 satellite offices.
- **Proactive Monitoring:** Developed service-aware alerting and telemetry frameworks for business-critical systems, reducing mean time to resolution (MTTR) through proactive incident detection and enhanced observability.
- **Cybersecurity Governance:** Directed enterprise-wide vulnerability scanning and remediation efforts for internal and external

systems to maintain a hardened security posture.

- **Enterprise Mobility:** Onboarded and managed 15,000 iPads via Meraki MDM, implementing automated security protocols including geofencing, remote-wipe triggers, and custom application deployment.

Systems Administrator

- **Operational Visibility:** Modernized infrastructure monitoring by designing custom Grafana dashboards to visualize real-time performance metrics and network telemetry.
- **Disaster Recovery:** Spearheaded the deployment of Datto backup and disaster recovery appliances to ensure 100% business continuity for critical operations.

SKILLS

Networking: BGP, DNS, VPN, VRRP, High Availability, F5 Load Balancing, Intrusion Detection, Vulnerability Scanning, Meraki MDM

Languages: Python, Bash, PowerShell, SQL, YAML, Javascript, PHP

Automation/Orchestration: Terraform, GitOps, CI/CD Pipelines, Ansible Automation Platform, REST API, GraphQL, Grafana, Middleware Development, Gitlab, ServiceNow, Unit Testing

Infrastructure: Infrastructure as Code (IaC), VMware vSphere, Check Point Firewalls, Azure Cloud Services, OS Hardening (CIS/STIG), Nutanix AHV, Rubrik, Active Directory, DBaaS, NUMA Tuning

EDUCATION

Bachelor Of Science in Computer Information Systems

12/2012

SUNY Fredonia

CERTIFICATIONS

Certified Meraki Network Operator

Meraki

VOLUNTEERING & LEADERSHIP

Habitat For Humanity

07/2025 - Present

Volunteer

Tift Nature Preserve

01/2025 - Present

Volunteer

716 Paws

01/2022 - Present

Thermal drone pilot for search and rescue missions.